

Data analytics for personalized genomics and precision medicine**Lecture8 scribing**

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Lecture Outcome

- Classification: Why? What? How?
- K-nearest neighbour classification
- Clustering VS classification

1. Recap from last lecture**1.1 Clustering**

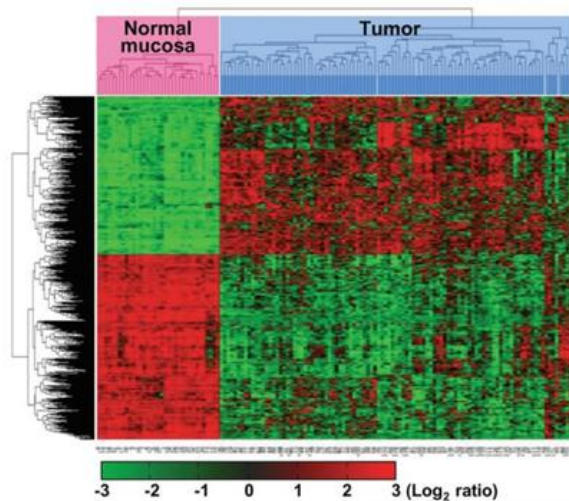
- Using similarity or dissimilarity to measure differences
- Correlation, Euclidean distance, Manhattan distance , Mahalanobis distance

1.2 Hierarchical clustering

- Producing a set of nested clusters
- Corresponding to meaningful taxonomies
- Briefing on the steps : similarity/distance matrix → merge the two closest clusters → update → merge again → repeat until a single cluster remains

2. Classification**2.1 Why?**

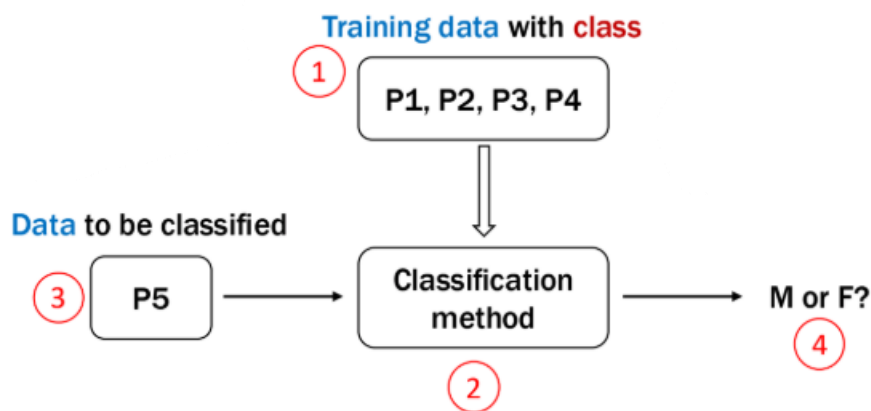
- Classifying items: better organization
- Classifying people: patients → corresponding treatment, customers → precise targeting
- Classification in biology: for gene expression e.g. tumor



2.2 What is it?

- Given a collection of records and each record has a set of attributes, one of them is the class → assign the class of previously unseen records based on attributes and training set as accurately as possible

2.3 How to perform classification



- Training data with class, classification method and data to be classified are needed

3. K-nearest neighbours classification

3.1 Definition

- An algorithm that stores all available instances and classifies new ones based on a distance metric to the available ones

- Training process → predicting (find K training stances that are closest to the query instance, return the most frequent class label) → normalized

3.2 What should be determined

- Distance metric
- Value of K (usually between 5 and 10 for most low-dimensional data set or chosen by using cross-validation)
- Weighting function (optional)

3.3 Procedure of KNN

1. Normalization

Person	Height(m)	Weight(kg)	Gender
P1	1.79	75	M
P2	1.64	54	F
P3	1.70	63	M
P4	1.88	78	M
P5	1.75	70	??

Person	Height	Weight	Gender
P1	0.625	0.875	M
P2	0	0	F
P3	0.25	0.375	M
P4	1	1	M
P5	0.4583	0.6667	??

2. Compute distances

Person	P5	Gender
P1	0.267	M
P2	0.809	F
P3	0.358	M
P4	0.636	M
P5	0	??

3. Identify the K most similar data

Person	P5	Gender
P1	0.267	M
P2	0.809	F
P3	0.358	M
P4	0.636	M
P5	0	??

4. Take their class out and find the mode class

4. Clustering VS Classification

	Clustering	Classification
Goal	Find similarity	Assign class to new data
Data	Without class	With class when training, without when testing
Classes	Unknown number	Known number
Output	Cluster index for each point	Class assignment of the testing data
Algorithm	One phase	Two phases(training and application)

Type of machine learning	Unsupervised	Supervised
Definition	Machine learning algorithms to analyse and cluster unlabelled data	Machine learning algorithms to classify and predict outcomes, trained on labelled data
Examples	Clustering, dimension reduction	Classification, regression

5. Useful links

KNN in Python:

<https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html>